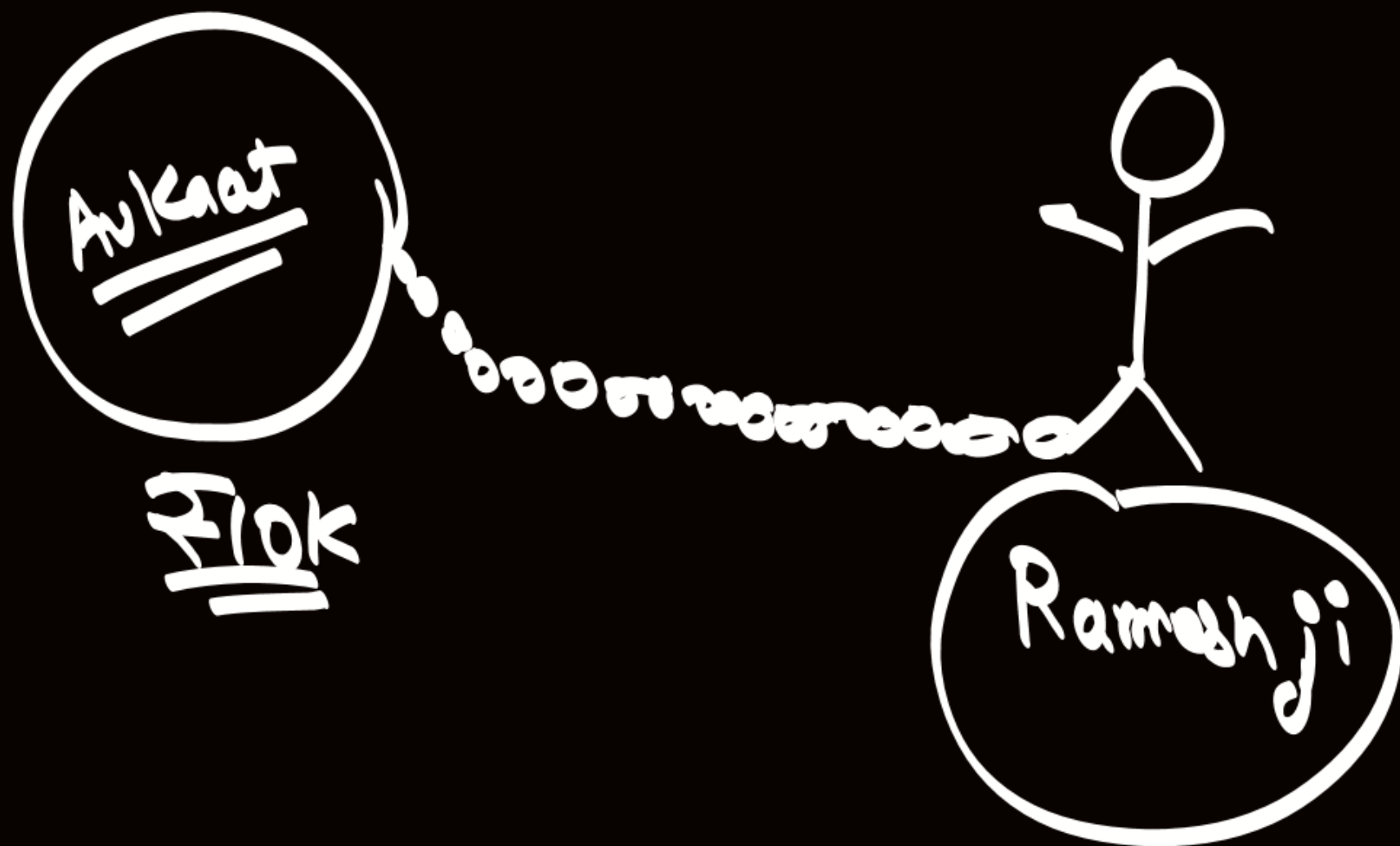




Unit : 1

Consumer Equilibrium

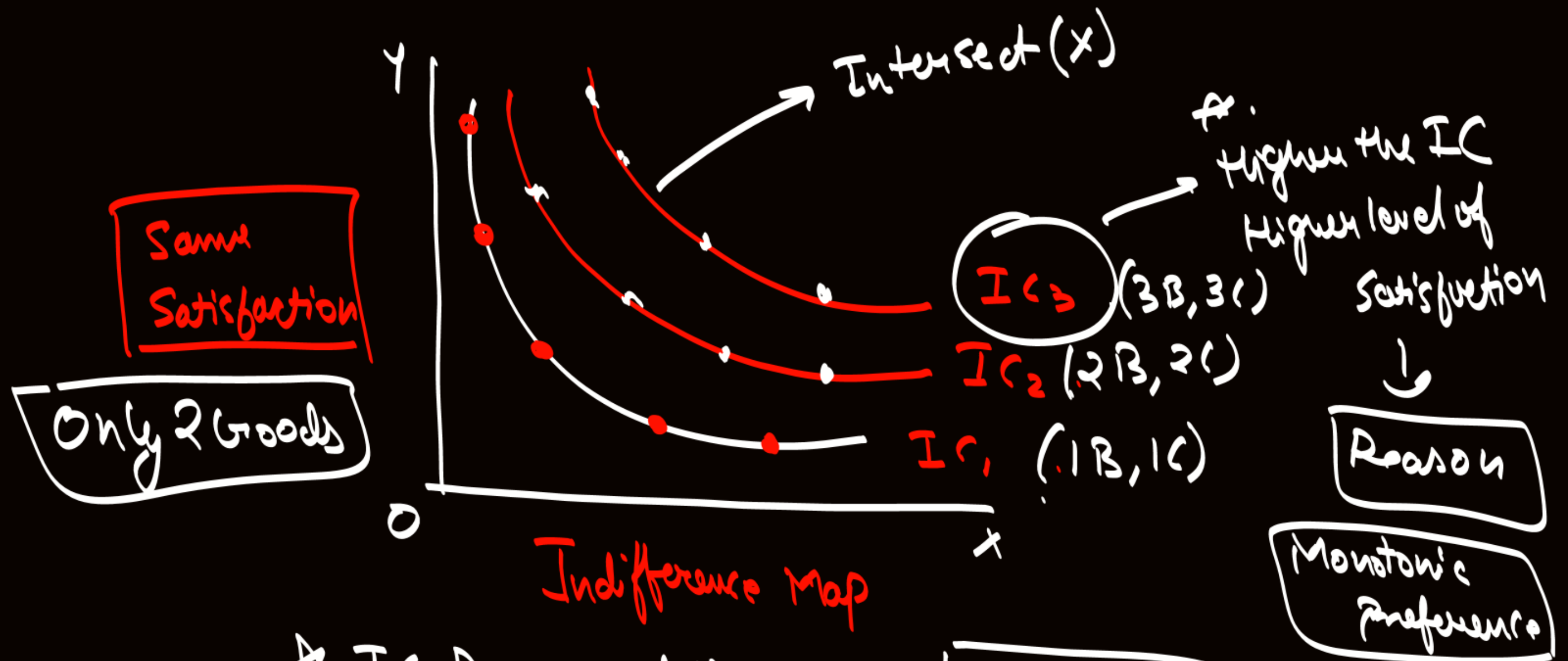
Chapter 2 | Lecture 7



BUGET LINE

= Income





- * IC Downward Sloping Curve: Inverse Relation
- * Convex Shaped: $MRS \downarrow$

Income = ₹20

Schedule

Apple ₹4 Banana ₹2

	Apple	Banana
A	5	0
B	4	2
C	3	4
D	2	6
E	1	8
F	0	10

₹20

₹20

₹20

₹20

₹20

₹20

A B C D E F G

Sets

Budget sets

combinations on points

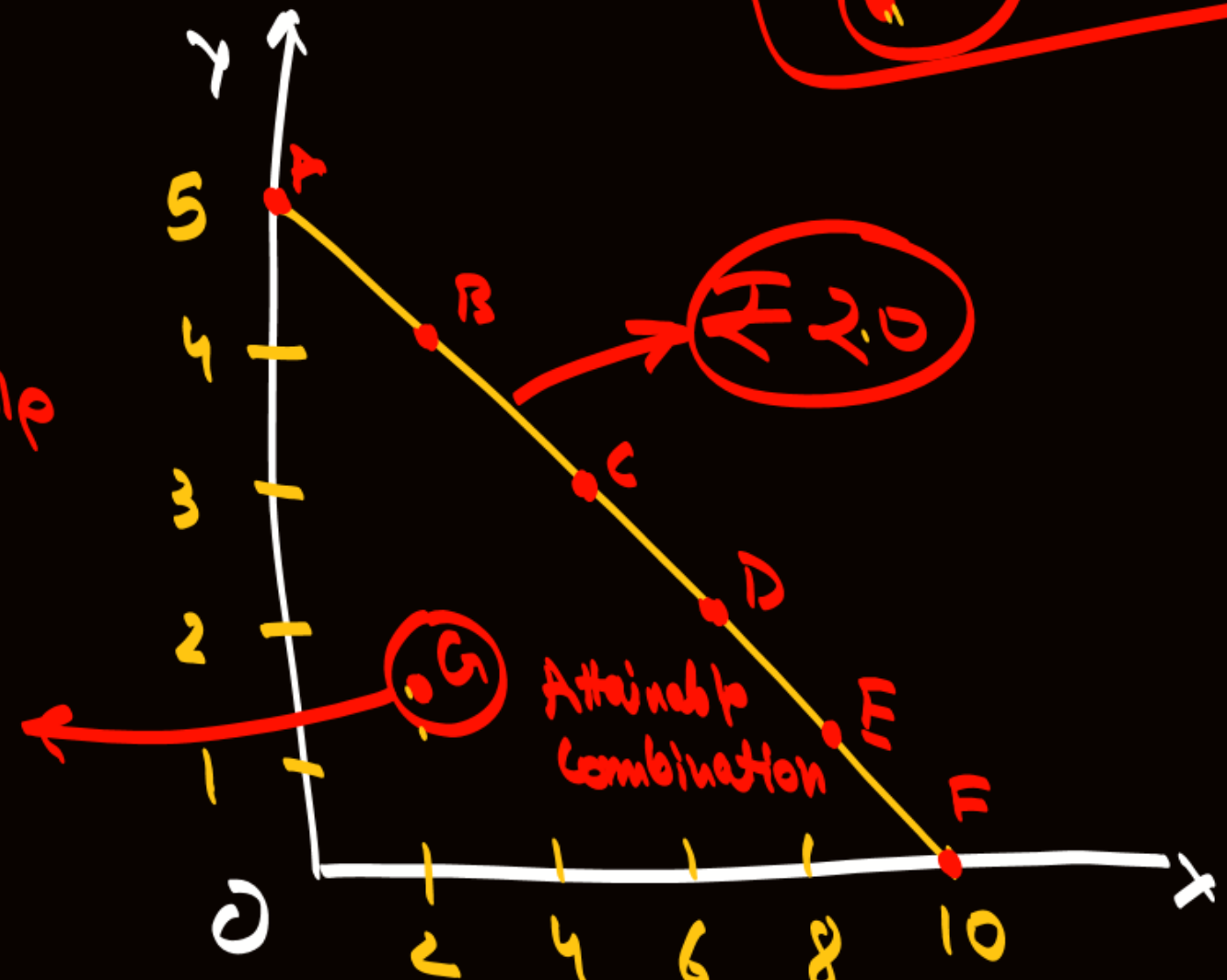
unattainable combination

H

₹40

Apple

₹10

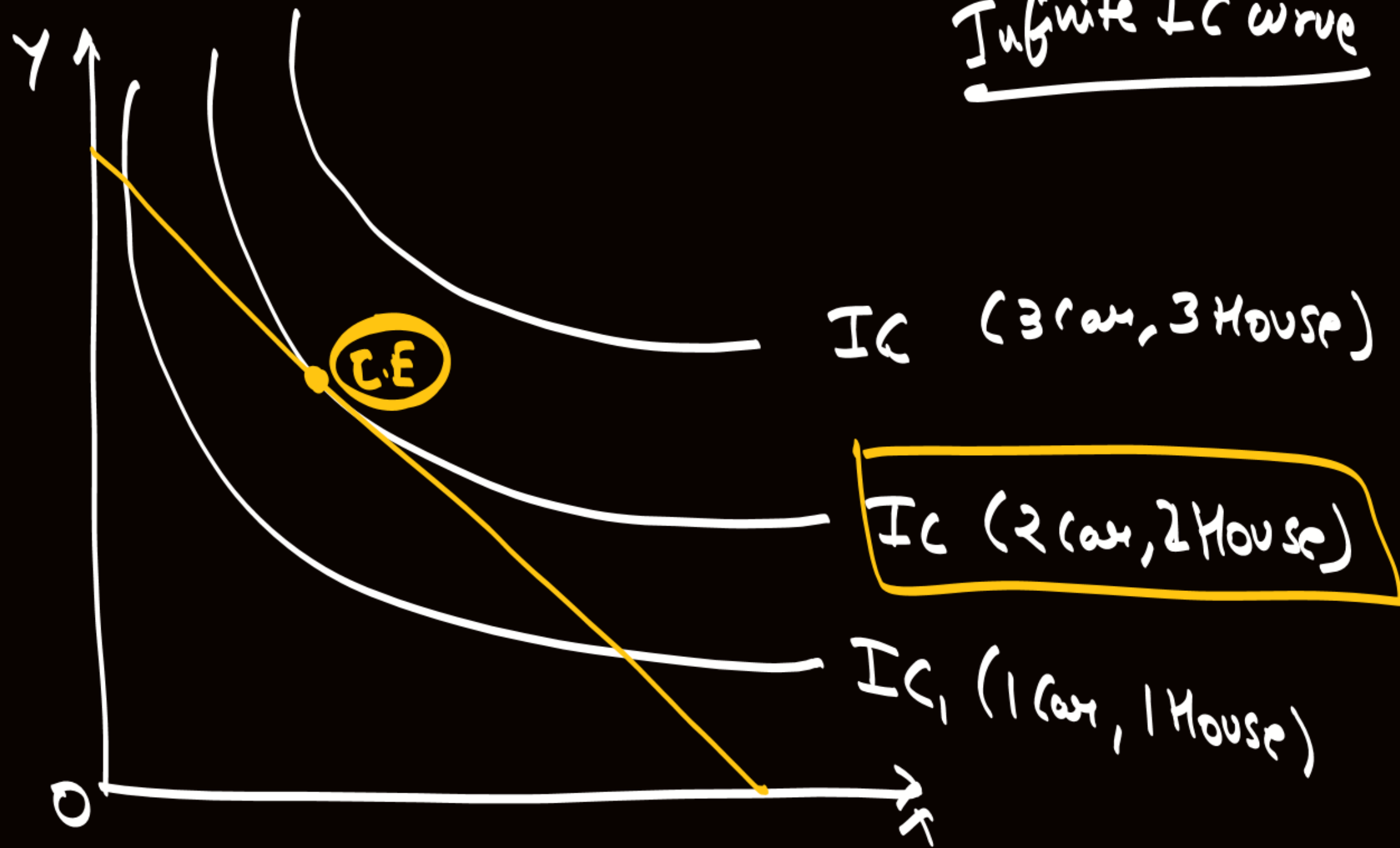


₹20

Attainable combination

Banana Budget line

Infinite IC curve



Budget line is a graphical representation of all possible combinations of two goods which can be purchased with given income and prices, such that the cost of each of these combinations is equal to the money income of the consumer.

Budget Line is also known as: (a) Price Line; (b) Price Opportunity Line; (c) Price-Income Line; or (d) Budget Constraint Line.



Suppose, a consumer has an income of 20. He wants to spend it on two commodities: X and Y and both are priced at 10 each. Now, the consumer has three options to spend his entire income: (i) Buy 2 units of X; (ii) Buy 2 units of Y; or (iii) Buy 1 unit of X and 1 unit of Y. It means, possible bundles can be: (2, 0); (0, 2) or (1, 1).

When all these three bundles are represented graphically, we get a downward sloping straight line, known as 'Budget Line'. It is also known as price line.



₹ 10



₹ 10

BUDGET SET

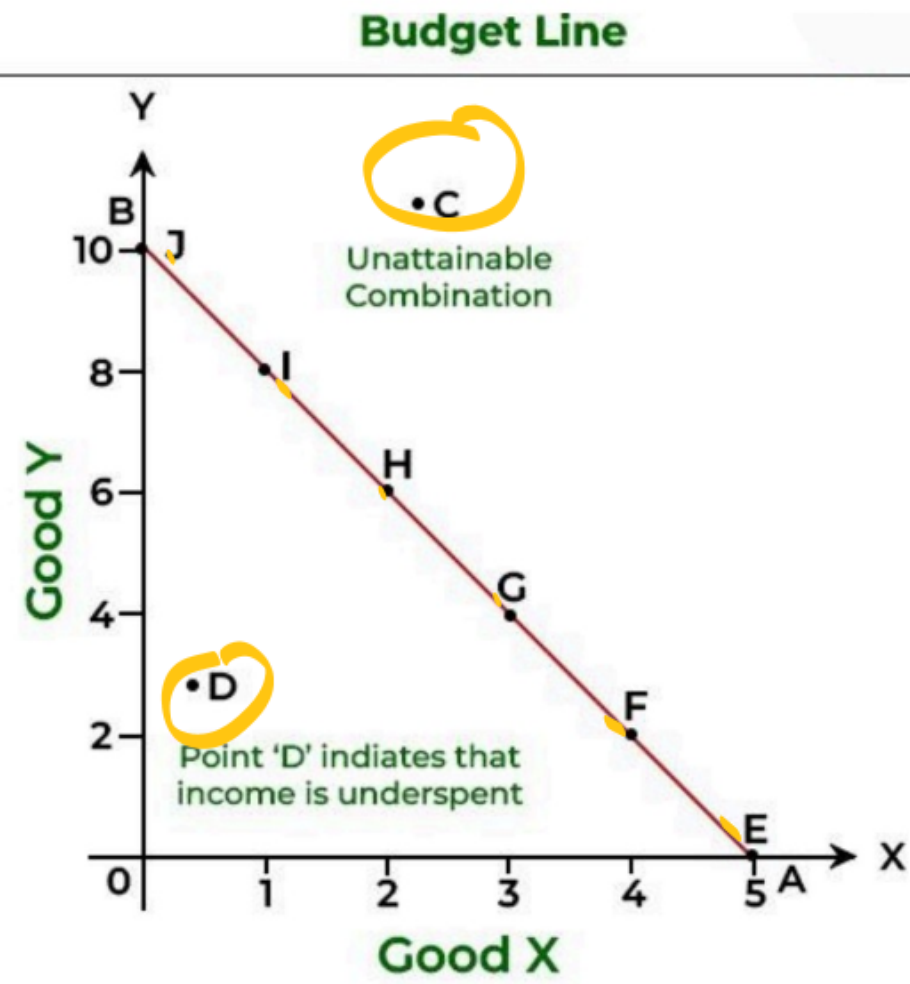
Budget set is the set of all possible combinations of the two goods which a consumer can afford, given his income and prices in the market.

In addition to the three options, there are some more options available to the consumer within his income, even if entire income is not spent. Budget set includes all the bundles with the total income of 20, i.e. possible bundles or Consumer's bundles are: (0, 0); (0, 1); (0, 2); (1, 0); (2, 0); (1, 1). Consumer's Bundle is a quantitative combination of two goods which can be purchased by a consumer from his given income.





Combination of Apples & Bananas	Apples (A) (₹ 4 each)	Banana (B) (₹ 2 each)	Money spent = Income (₹)
E	5	0	$(5 \times 4) + (0 \times 2) = 20$
F	4	2	$(4 \times 4) + (2 \times 2) = 20$
G	3	4	$(3 \times 4) + (4 \times 2) = 20$
H	2	6	$(2 \times 4) + (6 \times 2) = 20$
I	1	8	$(1 \times 4) + (8 \times 2) = 20$
J	0	10	$(0 \times 4) + (10 \times 2) = 20$



IMPORTANT POINTS ABOUT BUDGET LINE

1. Budget line AB slopes downwards as more of one good can be bought by decreasing some units of the other good.
2. Bundles which cost exactly equal to consumer's money income (like combinations E to J) lie on the budget line.
3. Bundles which cost less than consumer's money income (like combination D) shows under spending. They lie inside the budget line.
4. Bundles which cost more than consumer's money income (like combination C) are not available to the consumer. They lie outside the budget line.
5. Budget Line is relevant (as compared to Budget Set) to determine Consumer's Equilibrium as it is assumed that consumer spends the entire income.

Attainable

Unattainable

MRT ↑
Production



PPC

MRS ↓
Consume



IC

M R. Exchange

Purchase



Slope = Budget line

ALGEBRAIC EXPRESSION OF BUDGET LINE

Budget set is the set of all possible combinations of the two goods which a consumer can afford, given his income and prices in the market.

Where,

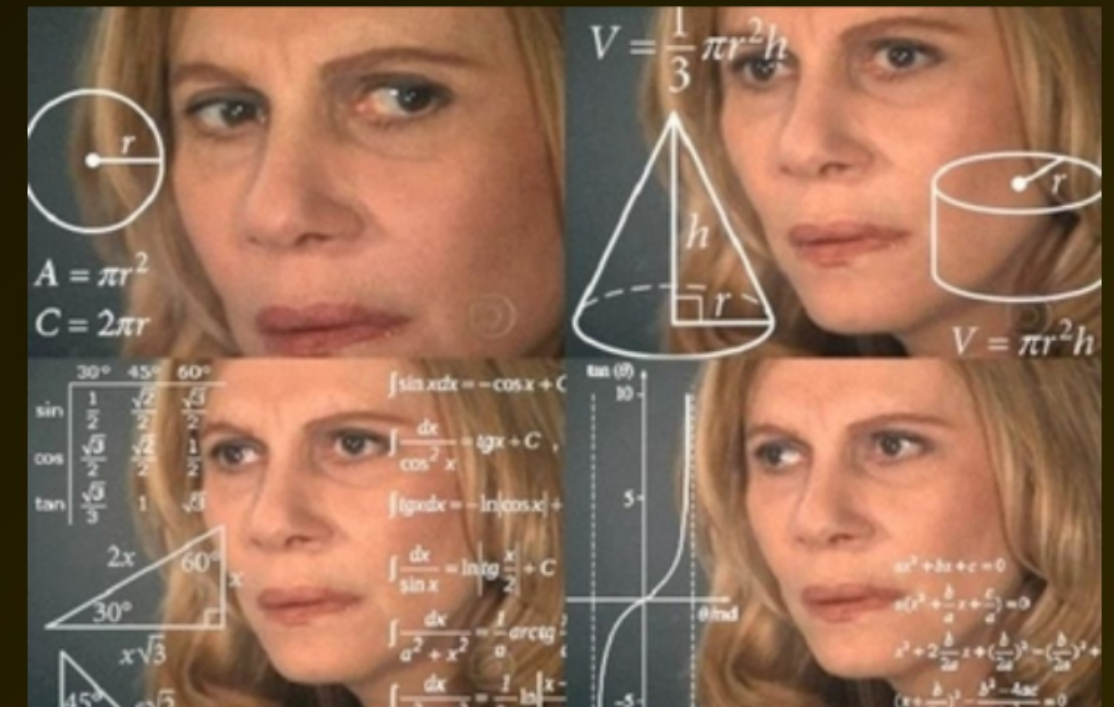
M = Money income;

Q_A = Quantity of apples (A);

Q_B = Quantity of bananas (B);

P_A = Price of apples (A);

P_B = Price of bananas (B);



$$M = (P_A \times Q_A) + (P_B \times Q_B)$$

$$\boxed{220} = (4 \times 20) + (2 \times 60)$$

Apple Unit Banana Unit

DIFFERENCE BETWEEN BUDGET LINE & BUDGET SET

SR.NO.	Budget Line	Budget Set
1	Budget Line include all the possible bundles that the consumer can purchase by spending his entire income at the given prices.	Budget Set include all the possible bundles which cost less than or equal to consumer's money income at the given prices.
2	The bundles of budget line lie only on the budget line.	The bundles of budget set lie either on or below the budget line.
3	Budget Line is expressed as: $M = (P_A \times Q_A) + (P_B \times Q_B)$	Budget Set is expressed as: $M \geq (P_A \times Q_A) + (P_B \times Q_B)$

SLOPE OF THE BUDGET LINE

- Slope of Budget Line = $\frac{\Delta A}{\Delta B}$
- Slope of Budget Line is denoted by Market Rate of Exchange (MRE)

$$\text{MRE} = \frac{\Delta \text{Sacrifice}}{\Delta \text{Gain}}$$

PROPERTIES OF BUDGET LINE

The two main properties of Budget line are:

- **Budget Line is Downward Sloping:** Budget line has a negative slope, i.e. it slopes downwards as more of one good can be bought by decreasing some units of the other good.
- **Budget Line is a straight line:** The slope of Budget line is represented by the Price Ratio. As Price Ratio is constant throughout, the budget line is a straight line.



$$\frac{4}{4}, \frac{4}{4}, \frac{4}{4}, \frac{4}{4}$$

$$\frac{1}{1} \text{ MRE}$$

Income = ₹200

Income = ₹100

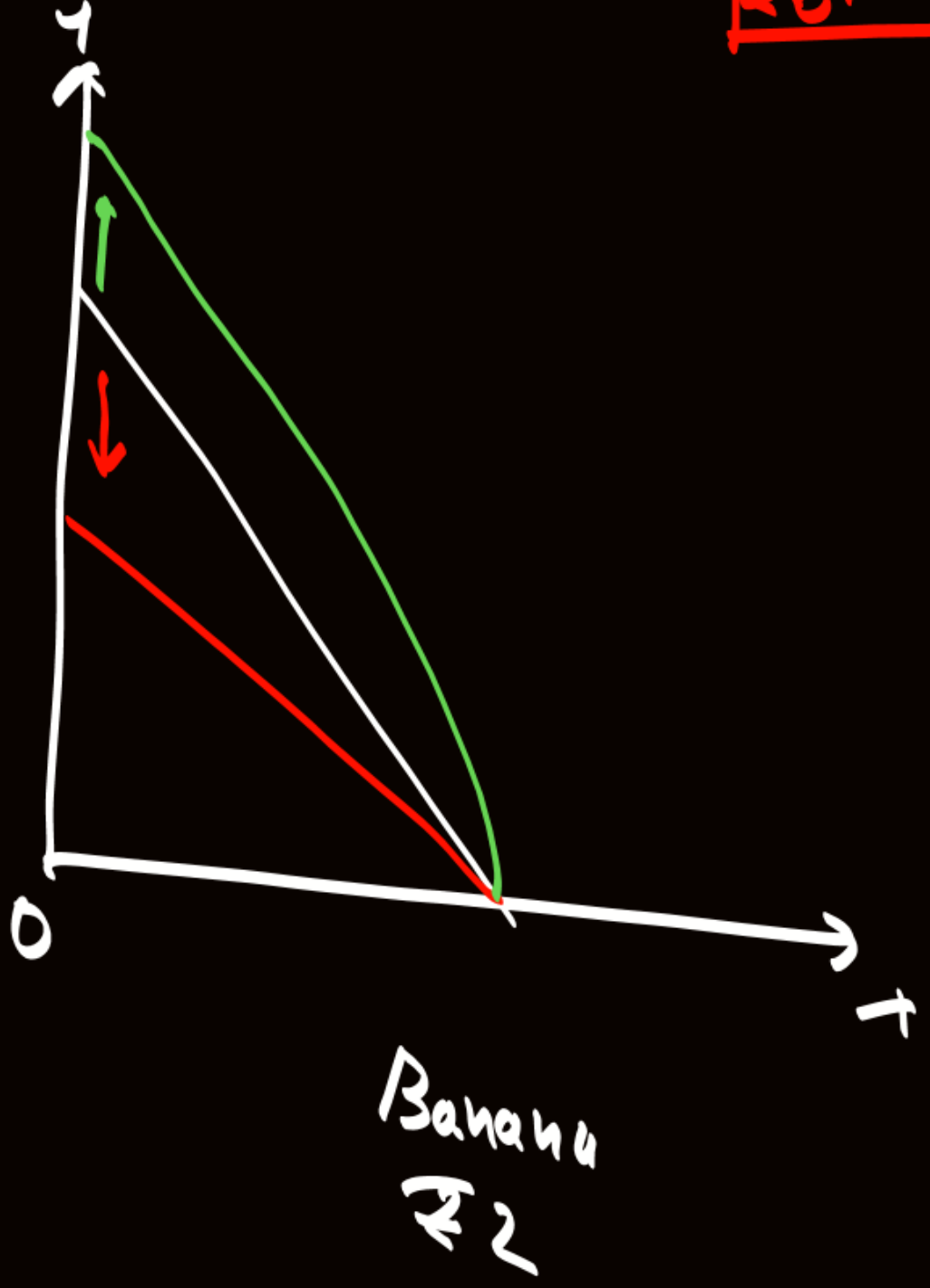
Income = ₹50



Shift in Budget line.

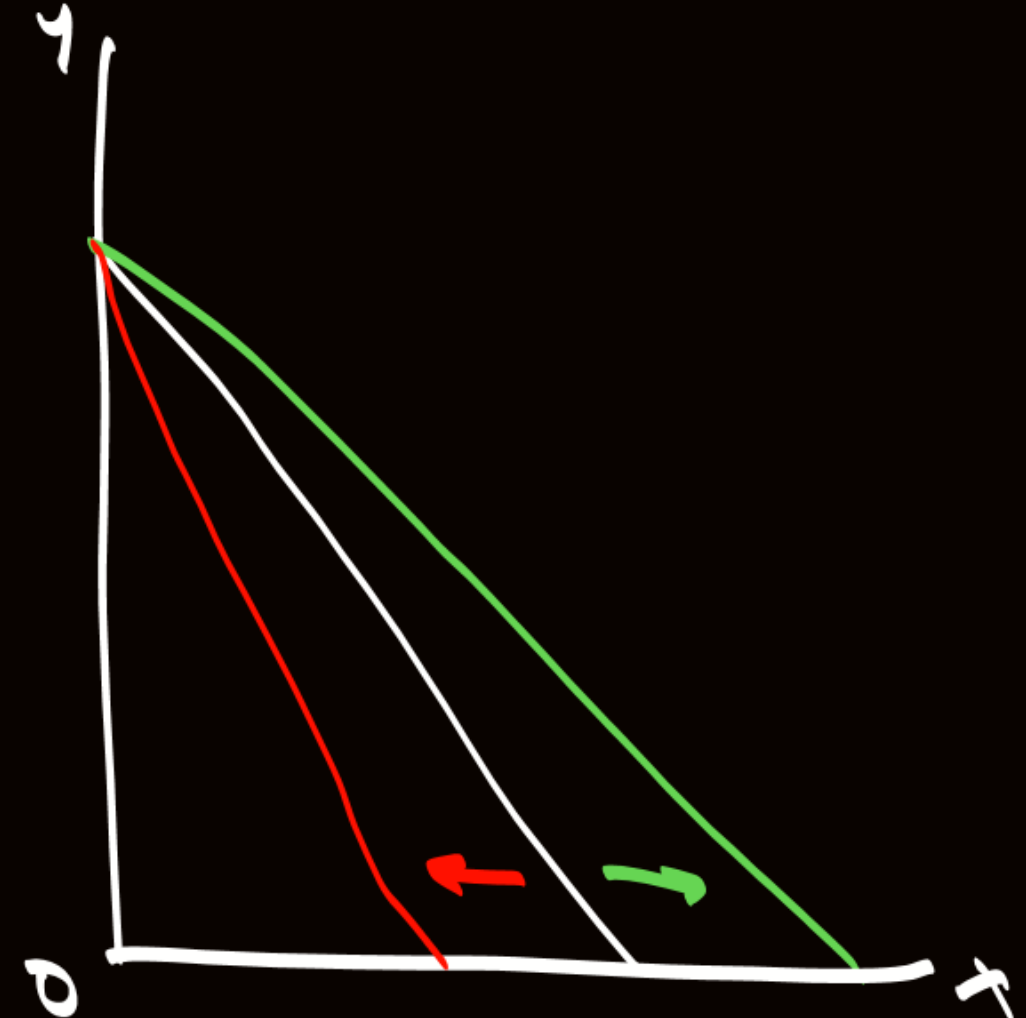
Rotation in Budget line

Apple = ₹2
 ₹4
 Apple = ₹6



Banana
 ₹2

Apple
 ₹4



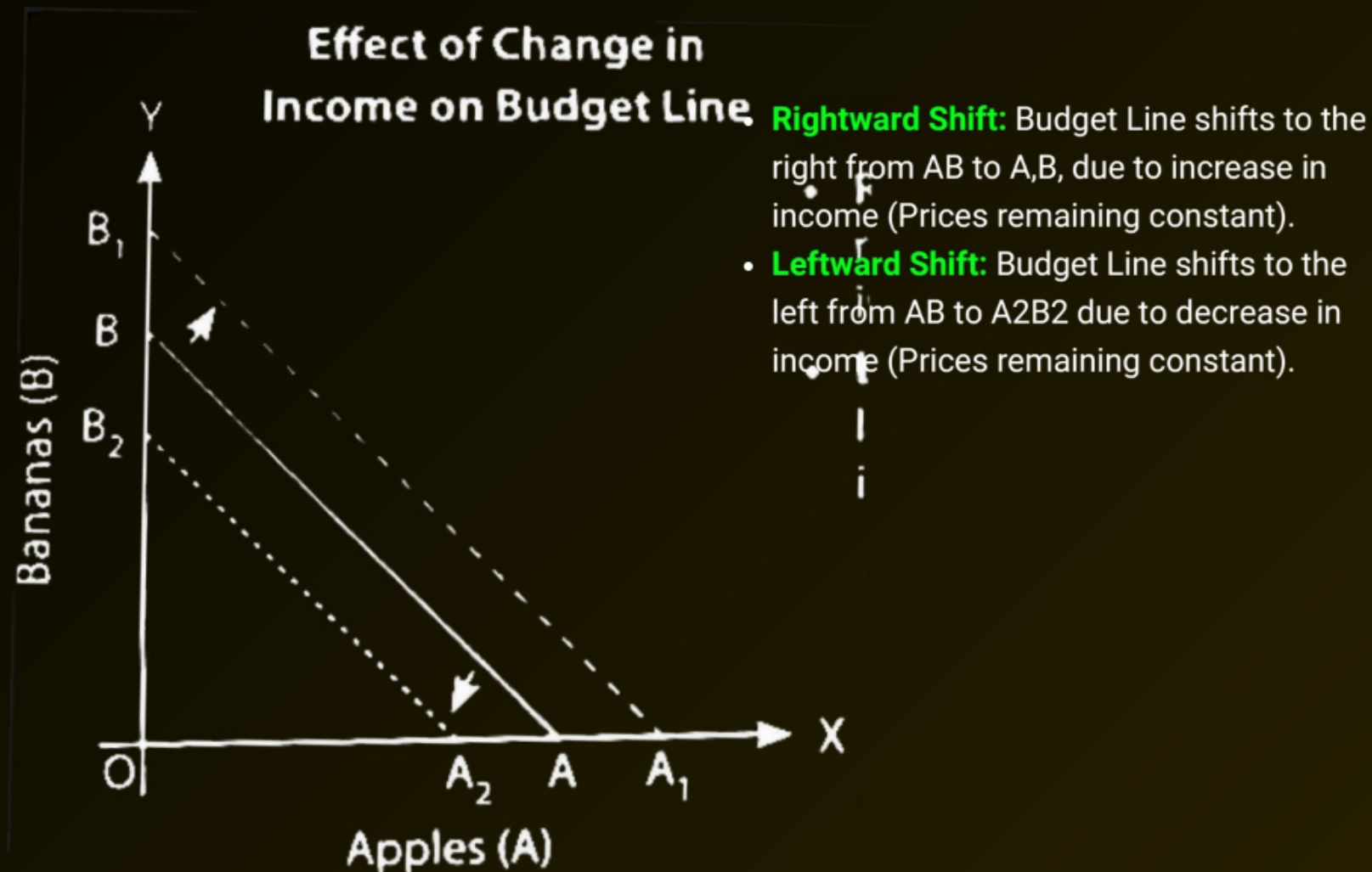
Apple
 ₹4

Banana
 ₹2

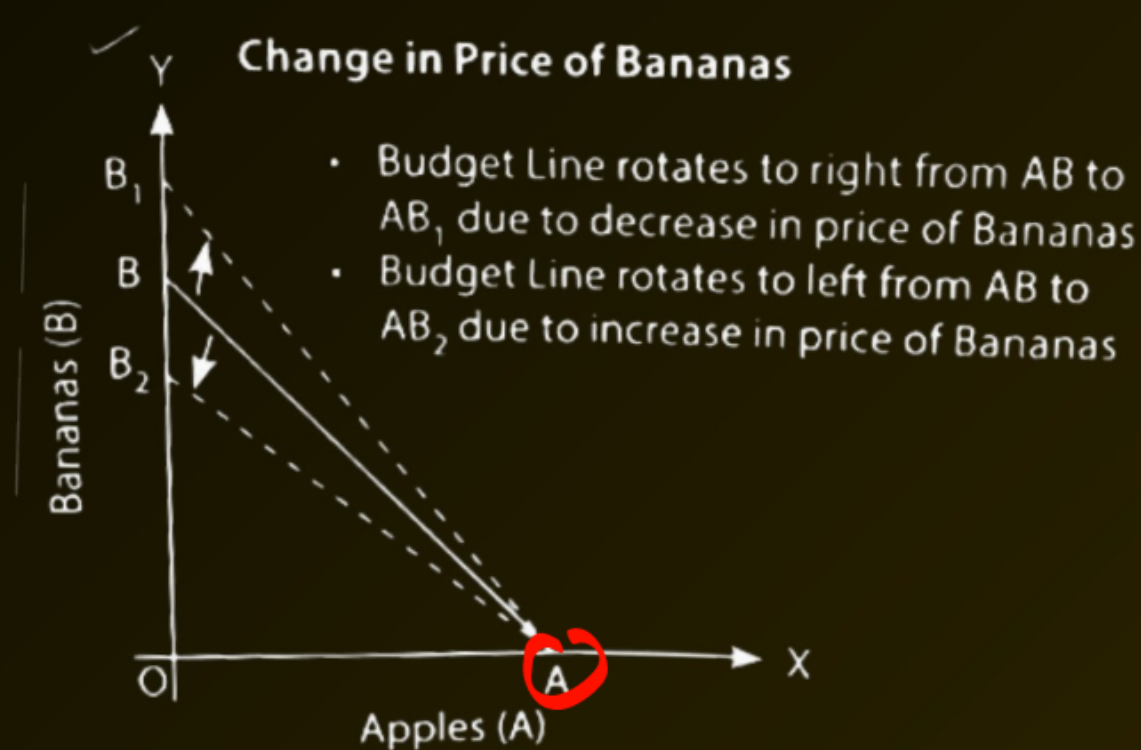
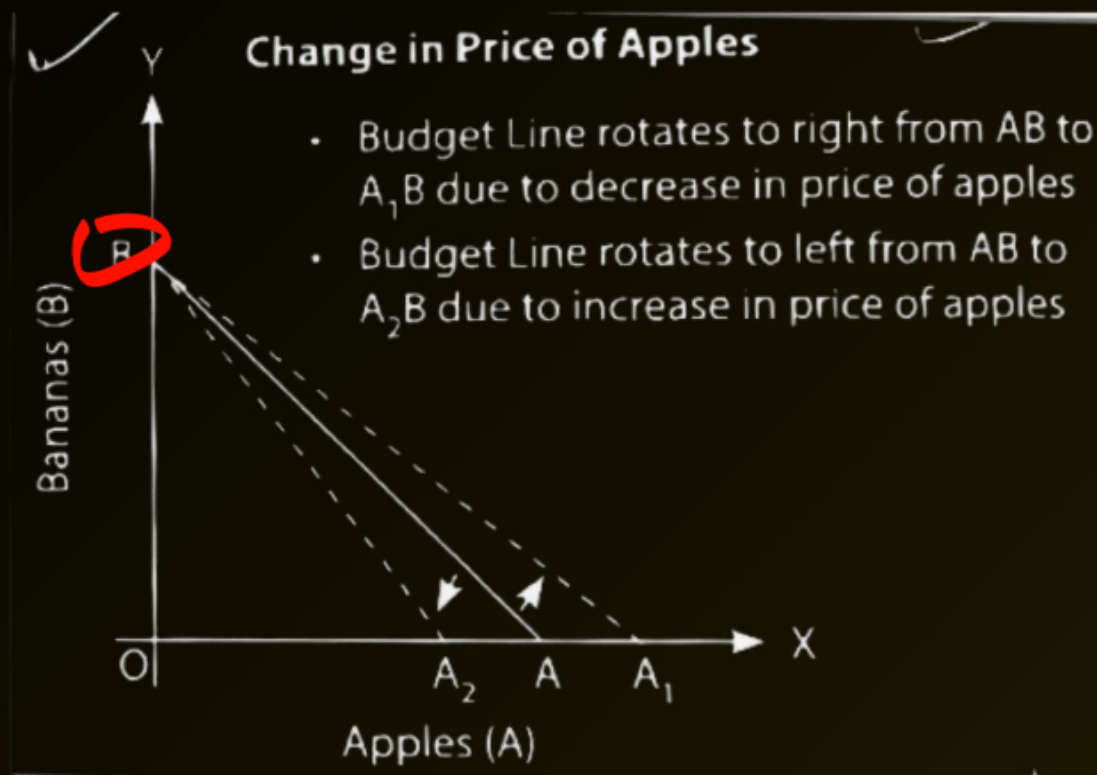
Banana
 = ₹1

SHIFT IN BUDGET LINE

- **Effect of a Change in the Income of Consumer:** If there is any change in the income, assuming no change in prices of apples and bananas, then the budget line will shift.



- Effect of change in Prices (Apples and Bananas):
 - Change in the Price of commodity on X-axis (Apples)
 - Change in the Price of commodity on Y-axis (Bananas)

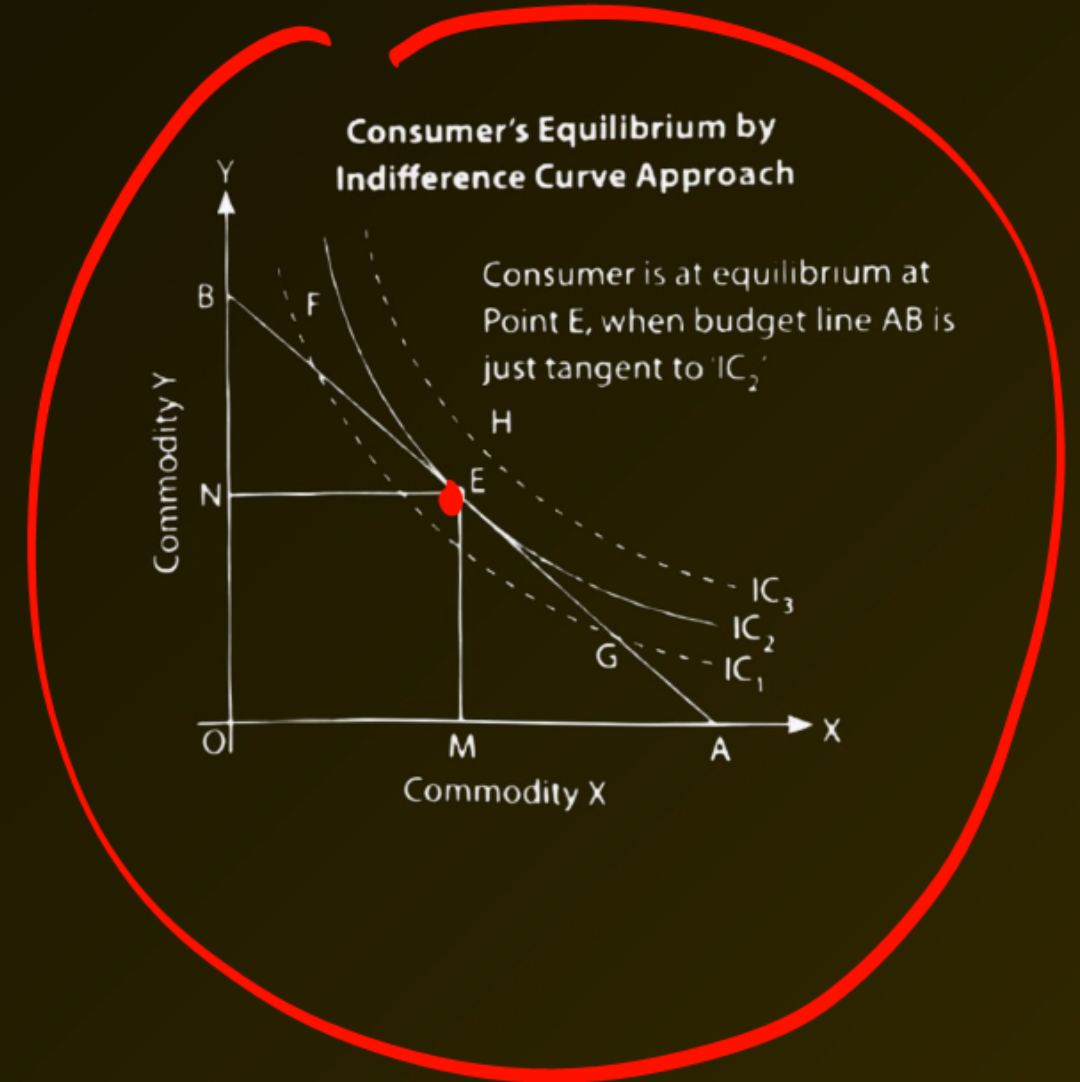


CONSUMER'S EQUILIBRIUM BY INDIFFERENCE CURVE ANALYSIS

The point of maximum satisfaction is achieved by studying indifference map and budget line together.

CONDITIONS

- MRS = Ratio of Prices or $\frac{P_x}{P_y}$ = Market Rate of Exchange (MRE)
- MRS continuously falls



ON TIME PRACTICE

OT P



WPP

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PARAKRAM

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